

1 Notation

We model four events each year: growth, colonization, fizzle and burnout.

Let (k, t) be the time index, where k represent the year index and t represent the season index, where $t = 0$ for the time at the beginning of the year, $t = 1$ for the time after growth, $t = 2$ for the time after colonization, $t = 3$ for the time after fizzle and $t = 4$ for the time after burnout. So $(k, 0)$ represents the same time as $(k - 1, 4)$

Assume there are n patches and we label them with p_i . For each patch we use $(\mathbb{1}_i^{(k,t)}, N_i^{(k,t)})$ to describe its state. $N_i^{(k,t)}$ represents population size in patch p_i at time point (k, t) , and $\mathbb{1}_i^{(k,t)}$ is defined as:

$$\mathbb{1}_i^{(k,t)} = \begin{cases} 1, & \text{if } p_i \text{ is infected at time } (k, t), \\ 0, & \text{otherwise} \end{cases}$$

Let $U \sim \text{Uniform}(0, 1)$ be a random variable uniformly distributed over $(0, 1)$, and we have

$$\mathbb{1}_{U \leq P} = \begin{cases} 1, & \text{if } U \leq P, \\ 0, & \text{otherwise.} \end{cases}$$

2 Growth

We can assume that reproduction of rats follows logistic growth, i.e.,

$$N_i^{(k,1)} = N_i^{(k,0)} + rN_i^{(k,0)}\left(1 - \frac{N_i^{(k,0)}}{K}\right)$$

where r is the growth rate and K is the carry capacity of a patch. We currently assume that all patches are of equal area and have the same carrying capacity.

3 Colonization

3.1 Population size

We assume that each individual rat move with probability c and land in a random patch. We ignore different distances and assume that all patches are of equal availability to a moving rat. So

$$N_i^{(k,2)} = (1 - c)N_i^{(k,1)} + \frac{c \sum_i N_i^{(k,1)}}{n}$$

The first term is population staying in the patch and the second term is population moving into the patch.

3.2 Infection

When an infected host dies the fleas are released and can travel between patches (such as being carried by a moving individual) and infect hosts in other patches.

Let z be the final size. The number of individuals that died of the disease in the previous year in a patch is $zDN_i^{(k-1,3)}$ if patch p_i is infected, and 0 if not. So the total number of death is $\sum_i zDN_i^{(k-1,3)} \mathbb{1}_i^{(k-1,3)}$.

Assume the probability of a susceptible patch getting infected is proportional to the number of death in the previous year, i.e.,

$$P_1^{(k)} = \frac{\alpha z D \sum_i N_i^{(k-1,3)} \mathbb{1}_i^{(k-1,3)}}{nK}$$

$\alpha \in [0, 1]$ is a scaling constant and nK is used to ensure $P_I^{(k)} \leq 1$

And

$$\mathbb{1}_i^{(k,2)} = 1 - (1 - \mathbb{1}_i^{(k,1)}) \mathbb{1}_{U < P_1^{(k)}}$$

4 Fizzle

Fizzle probability is

$$P_f = \frac{1}{\mathcal{R}_0}$$

And

$$\mathbb{1}_i^{(k,3)} = \mathbb{1}_i^{(k,2)} \mathbb{1}_{U < P_f}$$

Population size does not change.

5 Burnout

In infected patches, the population should be reduced:

$$N_i^{(k,4)} = (1 - zD \mathbb{1}_i^{(k,3)}) N_i^{(k,3)}$$

Burnout probability if a patch didn't fizzle is

$$P_b = \frac{P_1}{1 - P_f}$$

where P_1 is the probability of burnout. And

$$\mathbb{1}_i^{(k,4)} = \mathbb{1}_i^{(k,3)} \mathbb{1}_{U < P_b}$$